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Review on Additive Manufacturing of Single-Crystal Nickel-based Superalloys

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ABSTRACT

The conventional fabrication process for single-crystal nickel-based superalloy materials is directional solidification, which is classified as casting. With the rapid development of additive manufacturing (AM) technologies, a novel process for fabricating single-crystal superalloys has become possible. This article reviews recent research on the AM of single-crystal nickel-based superalloys. Laser AM technologies, particularly directed energy deposition, are mainly used to repair single-crystal materials. Electron beam powder bed fusion is an innovative method for the direct fabrication of single-crystal materials. Accordingly, the mechanisms of single-crystal formation during AM are analyzed to elucidate the potential of this process route. Furthermore, this article discusses the challenges faced by AM for single-crystal fabrication, and provides perspectives on the trends of future developments.

1. Introduction

Single crystal (SX) materials exhibit better high-temperature strength and creep resistance than polycrystalline materials. For instance, SX nickel-based superalloys have been produced to satisfy the requirements for aero-engines and gas-turbine blades [1]. In the 1970s, Pratt and Whitney first began research on SX superalloys. After a series of studies, PWA148 SX superalloy blades were successfully tested on a helicopter engine, PT6 [2]. Since then, SX superalloys have undergone four generations of development, with efforts from academia and industry. With the aid of protective coatings and cooling technology, the blades' operating temperature has been increased to as high as 1827 °C [3].

Similar to other superalloys, SX superalloys have characteristic microstructures mainly consisting of γ' phase and γ phase, and several other phases that precipitate during the forming process [4]. The γ matrix is a nickel-based austenite phase with a continuously distributed face-centered cubic structure, and usually contains a relatively large amount of solid solution elements (such as Co, Cr, Mo, and W). The γ' phase is an intermetallic compound that generally consists of Ni_3Al or Ni_3Ti . The γ' phase always precipitates in the γ matrix, and is the most important strengthening phase in nickel-based superalloys [5]. Its quantity, size, shape, and distribution have a decisive influence on the mechanical properties of the materials. After the fabrication of raw materials, delicate heat-treatment processes are often required to adjust

and control the shape and size of the strengthening phase [6]. For SX superalloys, there is a possible presence of a γ - γ' eutectic structure. The γ - γ' eutectic structure, with a randomized crystallographic orientation, makes the material a SX macroscopically; however, it is polycrystalline microscopically, and impairs the high-temperature performance [7].

Currently, there are four generations of SX superalloys; in each generation, the Re and Ru contents have gradually increased [3, 8]. A typical second-generation SX alloy, contains 3 wt.% Re. The Re content of third-generation SX alloys reached 6 wt.%. The fourth-generation SX alloys contain 6 wt.% Re and 3 wt.% Ru. An increase in Re can improve the high-temperature performance of the alloy, especially in suppressing the coarsening tendency of the γ phase at high temperatures. However, an excessive volume fraction of Re causes topologically close-packed (TCP) phases. Hence, platinum group elements, such as Ru, are generally added to control the stability of the microstructure.

Currently, the process applied for the fabrication of nickel-based superalloy SX parts, particularly turbine blades, is directional solidification casting [8]. The key issues in the SX fabrication processes are to inhibit the nucleation of stray grains (SGs) and to control the orientation of the grain during the solidification process. Nucleation can be inhibited by strictly controlling the temperature field, including the temperature gradient and solidification velocity. Thereby, the undercooling in front of the dendrites can be maintained at a low level, so that it is difficult for nucleation to occur. Generally, a Bridgman furnace is used to provide a stable and uniform temperature field.

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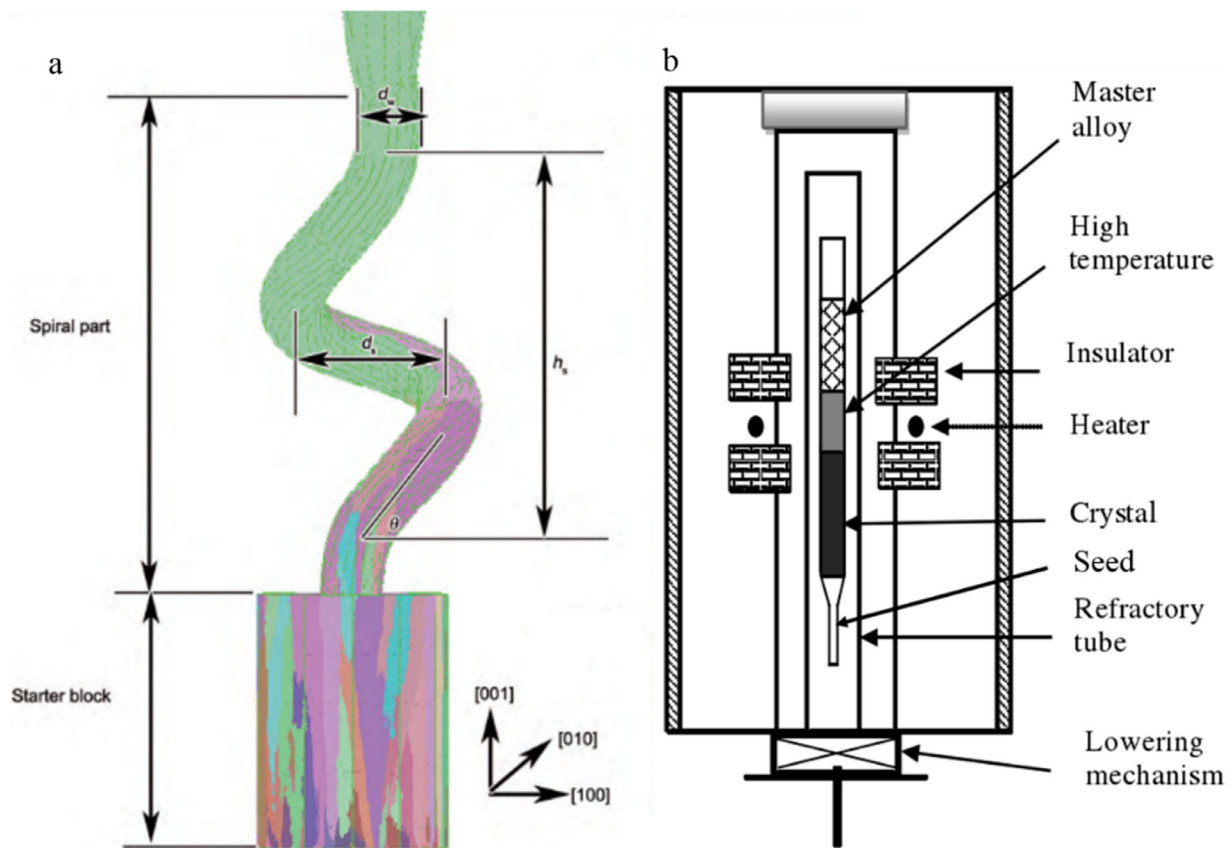


Fig. 1. Schematic diagram of (a) a spiral grain selector [3]; (b) a seed grain selector [9].

The grain selection and seed crystal technique are the two major techniques for controlling the orientation of the grain growth so as to obtain a SX. In the grain selection method, a spiral part named the spiral crystal selector is inserted between the starter block and the bottom of the parts shown in Fig. 1(a) [3], where h_s is the length of screw pitch, d_s is the diameter of spiral, θ is the initial angle of spiral, d_w is the diameter of helicoid. The spiral crystal selector geometrically stops most of the crystal grains from the starter block, and leaves only one crystal grain growing upward into the parts to form the final SX structure through dendrite expansion. The [001] orientation of the final SX is parallel or nearly parallel to the direction of the directional solidification device, but its [010] and [100] directions are randomly formed, and are difficult to control.

The seed crystal technique refers to a method where a SX of a specific orientation serving as a seed is installed at the bottom of a mold shell. The SX seed is partially melted after pouring, as shown in Fig. 1(b) [9]. The crystal grows along with the same orientation in the [001], [010], and [100] directions as the seed crystal, so as to obtain a SX with a controllable three-dimensional orientation. However, in the seed crystal method, miscellaneous crystals are often found at the contact position of the seed crystal re-melting zone and mold shell; these are caused by the extra nucleation driving force provided by the interface. Therefore, compared with the seed crystal technique, the spiral grain selection technique is more frequently used.

Besides directional solidification casting, additive manufacturing (AM) is another potential process for fabricating SX parts, although the capabilities have remained limited until now. AM technology, particularly metal AM technology, is gaining increasing attention from industry, owing to its superiority in direct fabrication of components with extremely complex metal structures (for example, turbine blades with inner cooling channels). Currently, many printing trials with nickel-based superalloys are being conducted by researchers, and samples and com-

ponents have been successfully fabricated. Almost all of these works are of a polycrystalline structure. Nevertheless, some works are pursuing a methodology of producing components of SX nickel-based superalloys.

In 2001, the SX structure shown in Fig. 2 was deposited on a SX substrate using the laser powder feeding method [10], also known as directed energy deposition (DED). This progress has demonstrated the potential for the repairment and even direct fabrication of SX blades with AM technology. Since then, aiming to repair SX blades, a large number of scholars have used laser heat sources to epitaxially grow SX structures on SX substrates. In this study, the fabrication of the SX structure was similar to that of the seed crystal method in casting.

Recently, breakthroughs were made in electron beam powder bed fusion (EB-PBF) technology. In 2016, Ramsperger et al. fabricated a CMSX-4 SX structure on a non-SX substrate [11]. Subsequently, Edouard et al. used EB-PBF technology to manufacture a SX bulk [12]. Li et al. fabricated larger-sized SX blocks [13]. These works further prove the extendibility and potential of AM in fabricating real SX superalloy parts. Evidently, metal AM technology is also based on controllable solidification. The key issues in fabricating SX nickel-based superalloys are the same as those in directional casting, i.e., inhabitation of the nucleation, and control of the grain orientation during the solidification. However, in AM technology, the material is deposited into a component in a layer-by-layer and track-by-track manner. The solidification process is quite different from that in directional casting. For instance, the temperature field is not stable and changes frequently, and the grain growth is not continuous, but rather is layer-by-layer. The mechanisms and methodology for fabricating SX nickel based superalloys should be more complex than those for directional casting. Nowadays, the methodology and mechanisms of the nucleation inhabitation, grain selection, and grain orientation control remain somewhat unclear, even though centimeter-scale SX bars have been successfully produced. For the purpose of promoting the research in the fabrication of SX nickel-based superalloys,

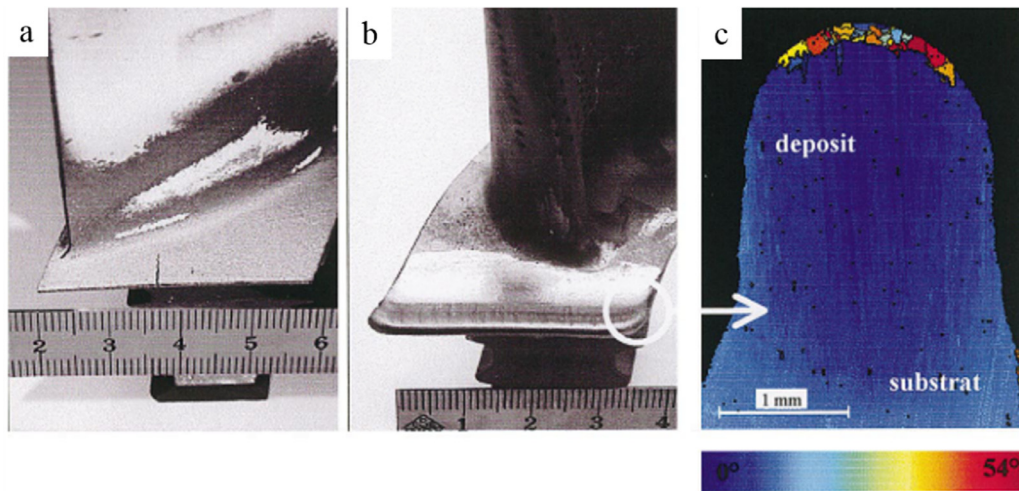


Fig. 2. Single-crystal (SX) repair of a damaged SX turbine blade: (a) cracked platform; (b) platform repaired by epitaxial laser metal forming; (c) electron backscatter diffraction grain-structure map of transverse section of repaired zone [10].

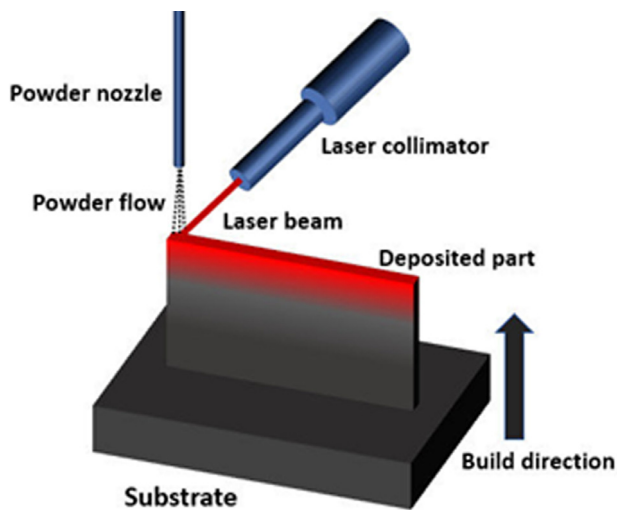


Fig. 3. Schematic illustration of directed energy deposition (DED) to build up a 3D part [14].

this article reviews the research progresses in the AM of SX nickel-based superalloys, including the repair of SX blades, fabrication of SX bars, and hypotheses for the mechanisms. The article also discusses the potential trends in this research area.

2. Fabrication of Superalloy Single Crystal by Laser Additive Manufacturing

Among the AM techniques using the laser as the energy source, the selective laser melting (SLM) technology has been widely used for a variety of materials. Nevertheless, the fabrication of nickel-based superalloy SX materials by SLM has seldom been reported in the existing literature. Currently, SX fabrications by laser AM are mostly achieved via the DED technique. DED uses high-energy beams to heat a substrate to form a local melt pool. Subsequently, a material transport system transfers the powder or wire to the formed melt pool. With the beam moving, the transported material is deposited on the substrate through a melting and solidification process. This procedure is continuously reciprocated to achieve the layer-by-layer formation of the materials. A schematic illustration of DED is provided in Fig. 3 [14]. The application scenario for the DED-prepared SX is the repair of SX turbine blades according to

the characteristic of this technique, i.e., the DED SX must be prepared on SX materials. By adjusting to the appropriate process parameters, the epitaxial growth of the SXs in the matrix is induced to form new SXs. With the advancements in research on the DED technique, critical issues regarding the fabrication of SX nickel-based superalloys have been profoundly investigated in the past decade, including the processing window, mechanism of columnar to equiaxed transition (CET), defect and crack formation during fabrication, anisotropy of the microstructure, and mechanical properties.

2.1. Laser AM process for superalloy SX

As an early attempt regarding SX repair/fabrication via AM, Gaumann et al. [10] completed the repair of small-size SX substrates. The successful experimental achievement resulted from their investigations regarding the influences of the nucleation undercooling, number of nucleation sites, and alloy composition on the CET [15]. They extended the CET model proposed by Hunt to a rapid solidification process based on the Kurz-Giovanla-Trivedi model, and studied the effects of the laser process parameters on the stability of the forming process, precision control of the formed parts, microstructure of the material, and performance of the laser metal deposition (LMD) process [16]. As long as the temperature gradient and dendrite growth rate are determined, a direct relationship between the process parameters and solidification microstructure can be established, as shown in Fig. 4 [10]. From this encouraging result, the simple conclusion can be drawn that effective control of the CET during the solidification process can be realized using DED techniques.

A series of subsequent investigations further explored the mechanisms of SX formation in the DED process. Vitek et al. [17] considered the temperature gradient at the solid-liquid interface and change of the solidification speed with position, optimized the above-mentioned model, and further improved the accuracy. By using the volume fraction of the SGs as the dependent variable for quantifying the growth of SXs, and using the process parameters (laser power, scanning speed) and substrate grain orientation as the independent variables, they found that the volume fraction of the local SGs is related to the crystallographic orientation of the substrate. Anderson et al. [18] subsequently introduced heat flow and fluid modeling into the model. Their calculations showed that under the action of the vertical axis heat flow, the SGs tend to form in the intersecting areas of different dendrites. This study confirmed the influence of different substrate orientations on the formation of impurities in crystals.

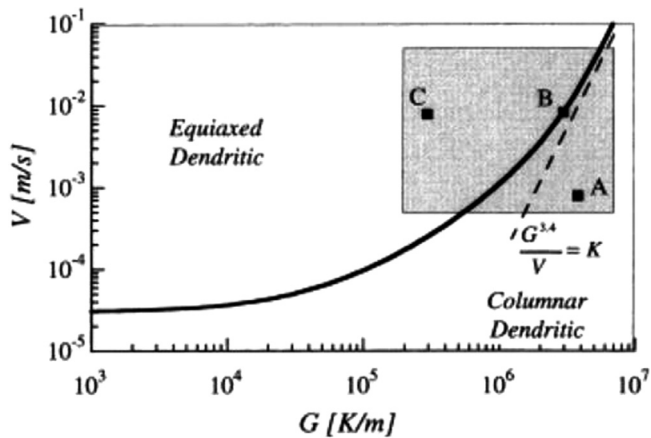


Fig. 4. Microstructure selection map for superalloy CMSX-4, showing the expected solidification morphology as a function of temperature gradient G and solidification velocity V [10].

Acharya et al. [19, 20] proposed a powder-bed-based AM process multi-physics coupling model, thereby revealing the influence of the powder on the geometry of the molten pool, size of the mushy zone, and location of the CET. Liu et al. [21] developed a multi-physics model that considered the powder feeding process, aiming to study the differences in microstructures under different substrate orientations. The simulation results showed that the powder feeding can change the location where the CET occurs, but has no significant effect on the overall crystal growth mode.

In addition to the modeling and simulation works, experimental efforts have been made. Liang et al. considered the influence of the powder feeding process on a SX epitaxial growth process window, and defined the feature ratio of the epitaxial SX growth as a quantitative standard for the microstructure of the workpiece in the study, as shown in Fig. 5 [22]. Buchbender et al. [23] investigated deposition strategies for extending the height of a SX, with quantitative consideration of the cracking and misorientation formations. They indicated that a continuous meandering strategy combined with an additional remelting step showed the most promising percentage of single-crystallinity, along with an increase in the epitaxial deposition height.

The above-mentioned studies deeply analyzed the solidification process in DED from the perspective of the processing details contributing to obtaining suitable process parameter windows for SX repair.

Xiao et al. [24] utilized a quasi-continuous-wave laser AM technique very analogous to a typical DED technique to build up a SX-like structure. By delicately adjusting the process parameters, the epitaxial growth of the columnar grains was strengthened, leading to the zigzag-oriented microstructures shown in Fig. 6. Even though the fabricated material was not an SX, this study undoubtedly provided direct proof that the formation of columnar grains is directly linked to the morphology of the melt pool, which affects the heat dissipation direction and thermal gradient; moreover, the arrangements and morphologies of the melt pools can be controlled according to the scan strategy and process parameter(s).

Zhou et al. [25] reported successful fabrications of SX CSU-B1 thin-wall samples by DED. However, the block samples suffered from perceptible cracks, owing to large misorientation angles formed during deposition. They further indicated that the solidification cracks can be attributed to the shrinkage cavities and high melting-point inter-dendritic carbides, whereas the liquitation cracks can be attributed to the low-melting-point compounds. As a matter of fact, the hot cracking problem is one of the most crucial barriers to SX fabrications via DED. The mechanisms of the formation of hot cracking were deeply investigated by Lu et al. [26]. Their findings revealed that the initiation sites of hot cracking are mostly at the high-angle grain boundaries and low-angle grain

boundaries, with a critical threshold of approximately 7° . The excessive dendrite coalescence under-cooling undermines the stability of the liquid film in the inter-dendritic region. Moreover, Re, an important alloying element in the third-generation superalloys for strengthening the γ' phase at high temperatures, has been found to be prone to being precipitated in the central inter-dendritic region (shown in Fig. 7). Hence, the liquid feeding is inhibited by the Re-rich precipitates, promoting the formation of hot cracks.

2.2. Laser AM microstructures for superalloy SX

The microstructures of AM superalloys share some similarities with those of their cast/wrought counterparts. In the SX state, the microstructure consists of one single grain filled by dendrites taking the same orientation. However, the morphologies of the dendrites in a conventionally fabricated substrate and an AM sample are quite different. As shown in Fig. 8(a) and 8(b), finer dendrites can be seen in the DED sample relative to the as-cast substrate. The solidification rate and thermal gradient in DED are much higher than that in casting. Similar to the Hall-Petch law in the grain-refining effect, finer dendrites can contribute to the strengthening of materials.

As mentioned above, the hot-cracking problem remains a severe issue for the DED technique. In fact, the excessive solidification rate results in a high susceptibility to hot cracking. In terms of laser AM, SLM has a faster solidification process compared to DED. The thermal gradient in SLM can reach $\sim 10^6$ K/m. Thus, the solidification rate is much higher. SX fabrication by SLM has been tested by Yang et al. [27]. Even though very fine dendrites can be obtained via SLM as shown in Fig. 8(c), SLM is currently immature for SX superalloy fabrications, as the SX crack-free structure has a height of only approximately 2 mm. When the deposited height reaches 6 mm, the misorientation angle notably increases, and a crack is initiated. This finding also explains why DED is more appropriate for SX fabrication among the laser-based AM techniques.

Knowing that the solidification process of an AM superalloy has differences from that of conventional casting, the strengthening phase in its microstructure is worthy of attention. In the as-cast alloy, a large amount of γ' phase-forming elements, such as Al, Ti, and Ta, are often enriched between the dendrites, thereby forming a large amount of a eutectic and coarse γ' phase, while W and other γ phase-forming elements are enriched on the dendrite stem. This severe dendrite segregation results in an inhomogeneous distribution of the strengthening phases. With AM, the phenomenon of dendrite segregation has been alleviated, but not eliminated. The strengthening γ' phases in the as-deposited, solution heat-treated, and solution-aged heat-treated DD98M are presented in Fig. 9 [29]. The fine γ' phase is uniformly distributed on the γ matrix after the solution treatment, thereby eliminating the element segregation in the deposited state. After the solution-aging treatment, the size of the γ' phase is smaller than the γ' phase on the dendrite stem in the deposited state. The γ' phase shape has become more regular, with a cubic appearance, and is uniformly distributed on the γ matrix.

A post-heat-treatment for a repaired SX was investigated by Chen et al. [30]. The fast-cooling phenomenon in the AM leads to the low stability of the γ' precipitate in high-temperature service, as the precipitated γ' phases are very small in size, and have rounded morphologies. However, the standard heat treatment (HT) for alteration of the γ' phases is not applicable to the repaired SX. The high density of dislocations formed in the heat-affected zone is prone to recrystallization in a solution HT. Thus, they conceived a novel treatment as shown in Fig. 10, aiming to prompt the growth of γ' while keeping the AM superalloy in the SX state.

2.3. Laser AM mechanical properties for superalloy SX

The mechanical properties of DED SX superalloys have rarely been reported in literature. One of the reasons is that the preparation of

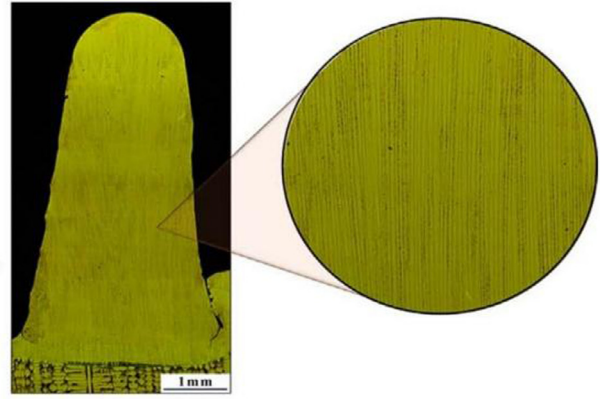
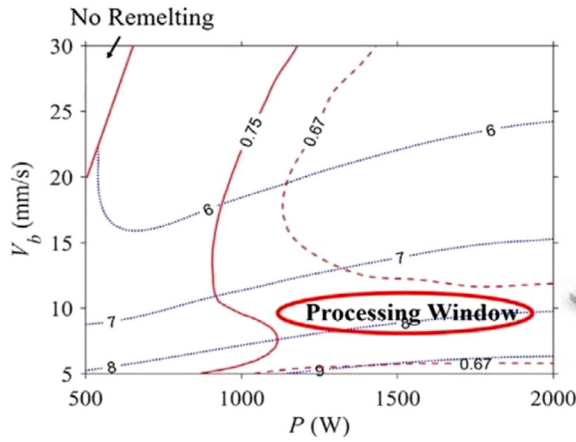


Fig. 5. Processing maps showing the influence of the power and scanning speed on remelting for different powder feeding rates, and a photo of the corresponding fabricated SX [20].

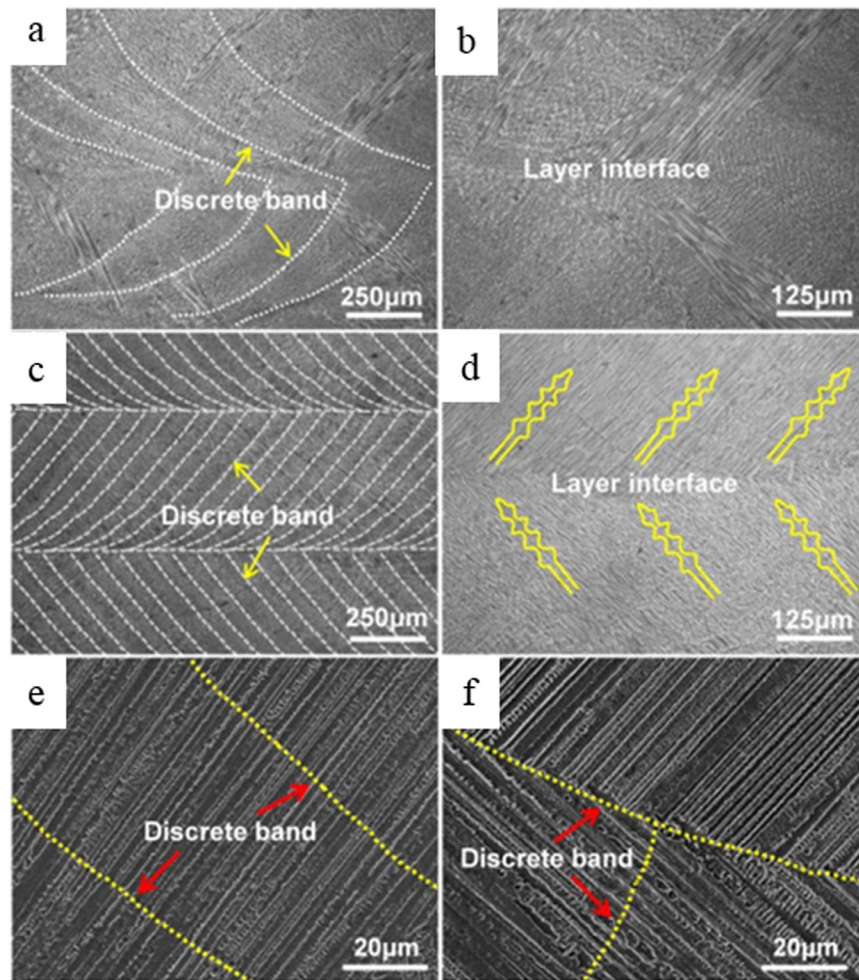


Fig. 6. Microstructures of as-fabricated samples: (a, b) optical microscope (OM) images of the continuous-wave (CW) sample; (c, d) OM images of the quasi-CW (QCW)-100 sample; (e, f) Scanning electron microscope (SEM) images of the QCW-100 sample [24].

a specimen-sized material block is temporally difficult. One inspiring work from Ci et al. [27] revealed certain DED characteristic mechanisms of the SX mechanical properties. They found that the high-temperature mechanical properties of DED SXs are strongly dependent on the anisotropy of precipitated carbides and γ - γ' eutectic, as well as the porosity. At moderate temperatures, the fragmentation of the coarse MC carbides in the substrate leads to cracking until final failure. At high temperatures, the tensile fracture is mainly owing to the γ - γ' eutectic in

the deposition. The difference in the fracture mechanisms reveals the characteristics of the microstructure of the DED SX. The fast solidification in the deposited area results in a carbide precipitation that is small in size in the inter-dendritic area. In the first several deposited layers, the cooling rate is very high, which inhibits the formation of the eutectic. As the cooling rate continuously reduces because of the repetitive successive laser heating, an increasing amount of eutectic. Schematic illustrations of the fracture mechanisms of a DED DD32 SX are provided

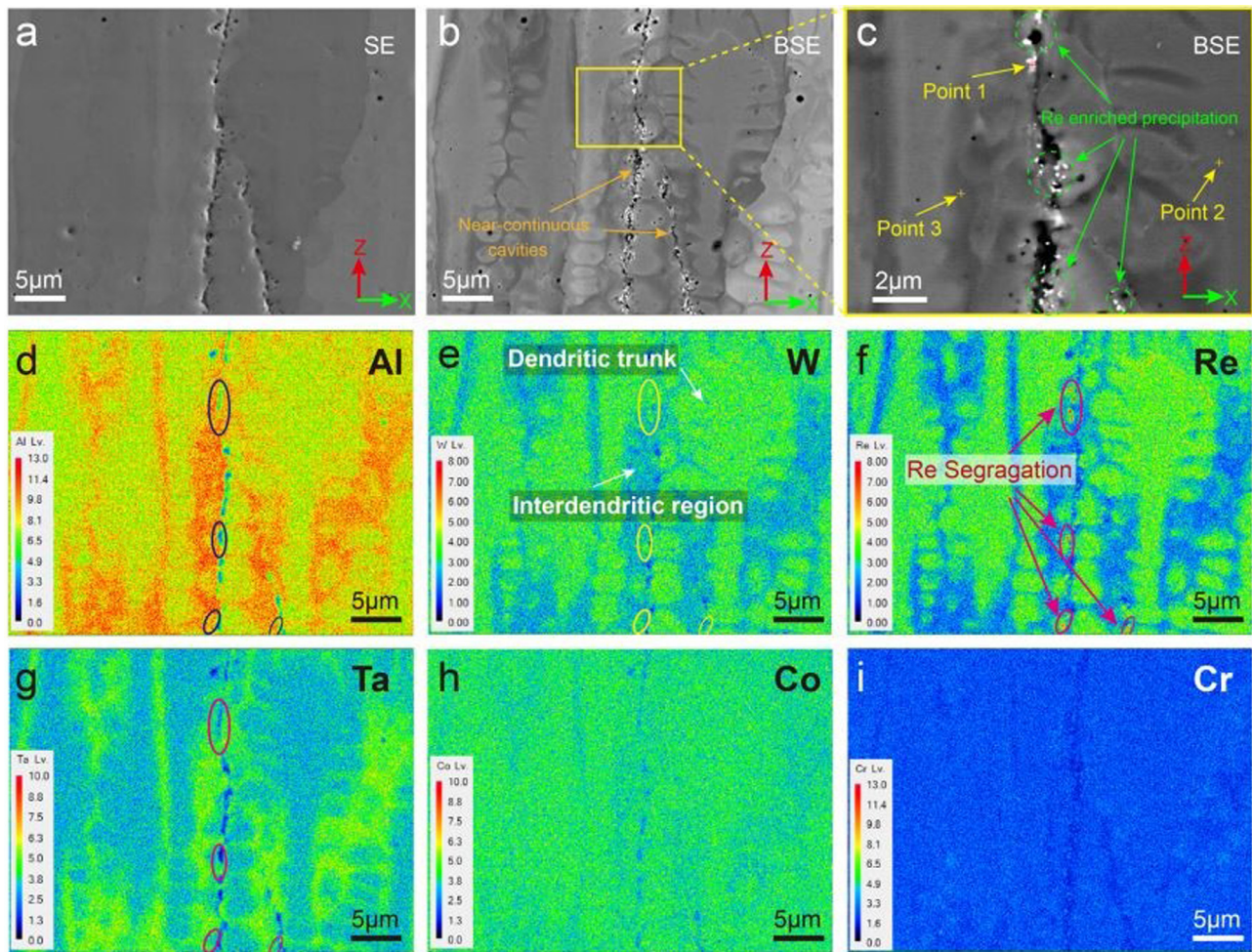


Fig. 7. Electron probe microanalysis (EPMA) elemental distributions of the region near the central crack: (a) SEM image of the crack initiation; (b) backscattered electron images of the crack initiation; (c) enlargement of the selected region in (b); and elemental distribution maps of: (d) Al; (e) W; (f) Re; (g) Ta; (h) Co; (i) Cr [26].

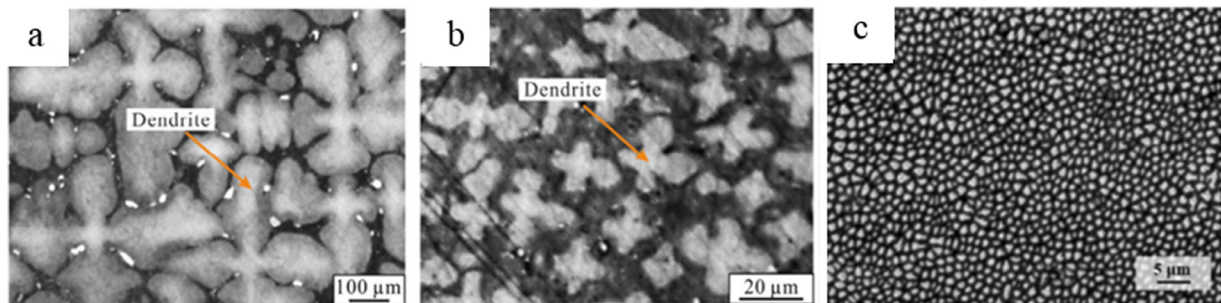


Fig. 8. Horizontal sections (perpendicular to the building direction) of the microstructures of (a) cast substrate [28]; (b) DED sample [28]; (c) SLM sample [27].

in Fig. 11. In addition to tensile tests, creep tests were performed on the same material, so as to compare with a cast material. The stress rupture lives of AM samples are longer than those of cast samples, with a lower elongation rate. This improvement is principally owing to the absence of the TCP phase and the uniform distribution, as combined with the proper size of the carbide precipitations.

To summarize, an AM method using a laser as the energy source can successfully complete the repair of SX superalloys. However, in the studies reported so far, because the temperature gradient and solidification rate at the top of the sample are lower than those at the bottom of the molten pool, stray grains can always be observed. In addition, the SX samples in the literature are small in size (mostly are just several layers thick), and are strictly limited to deposition on SX substrates. Thus, this

technology still focuses on repair, rather than on direct fabrication. Most studies simply report the microstructure, defect control, and method for obtaining the SX; the mechanical properties are rarely reported.

3. Fabrication of Superalloy SX by Electron Beam Powder Bed Fusion

As PBF AM technologies compared with SLM technology, EB-PBF technology uses an electron beam as a heat source; this can achieve a higher scanning speed, higher energy absorption, higher forming efficiency, lower thermal stress in the formed part, and higher preheating temperature [31–36]. A schematic illustration of the EB-PBF approach

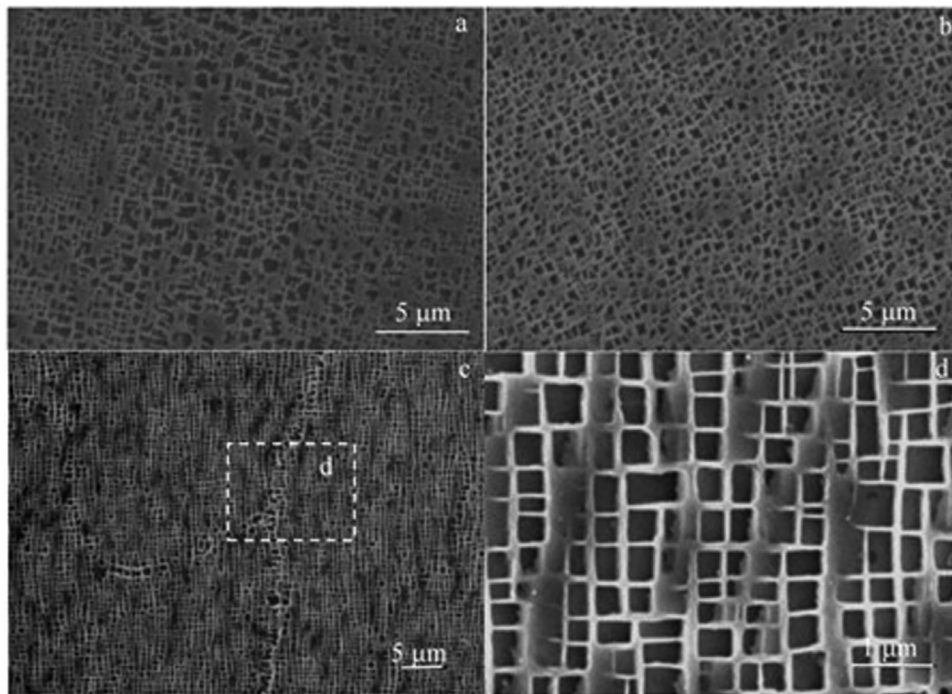


Fig. 9. Microstructures in different states: (a) deposition; (b) solution treatment; and (c, d) solution-aging treatment [29].

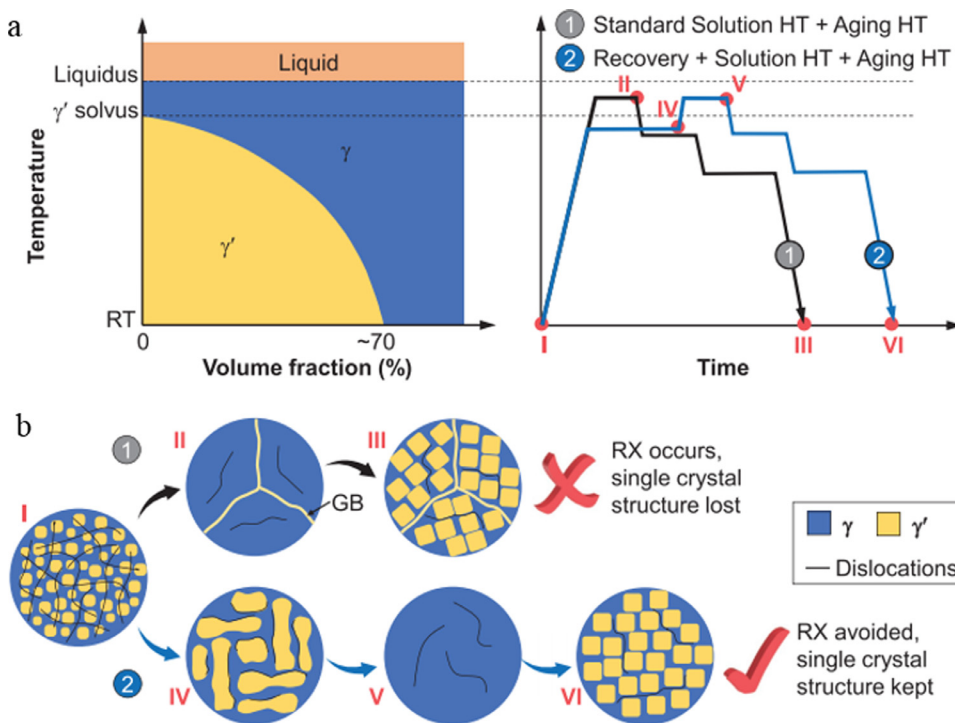


Fig. 10. Microstructural evolution of 3D-printed Ni-based single crystals in two different heat treatment (HT) procedures: (a) demonstrations of step-by-step HT with corresponding temperature intervals; (b) schematic illustrations of evolutions of the matrix, strengthening phase, and grain boundaries in the different HTs [30].

is shown in Fig. 12 [37]. EB-PBF technology has developed rapidly over the past 20 years. In recent years, SX samples have been successfully fabricated without the need for SX seeds and spiral grain selector structures firstly by the group led by Prof. Carolin Körner, who made great contributions to this topic.

3.1. Electron beam powder bed fusion process for superalloy SX

Ramsperger et al. [11] was the first to use EB-PBF technology to fabricate a SX sample without a grain selector or a SX seed as shown in Fig. 13(a). The SX was in the core of the sample above the grain selec-

tor, with a diameter of 8 mm and a 2 mm polycrystalline cortex. They disclosed that although the non-weldable nickel-based superalloy represented by CMSX-4 is highly sensitive to crack formation during the EB-PBF process, it is still possible to prepare the SX using a designed triangular structure in the bottom of the specimens for the EB-PBF. Moreover, compared with an SX obtained by casting, an SX obtained by the EB-PBF process has a finer and more uniform dendritic structure, and a lower fraction of the TCP phase. The research of Edouard et al. [12] used the EB-PBF process to prepare SXs through strict control of process parameters and process strategies, but without a designed triangular structure like Ref. [11] in the bottom of the specimens, as shown in Fig. 13(b).

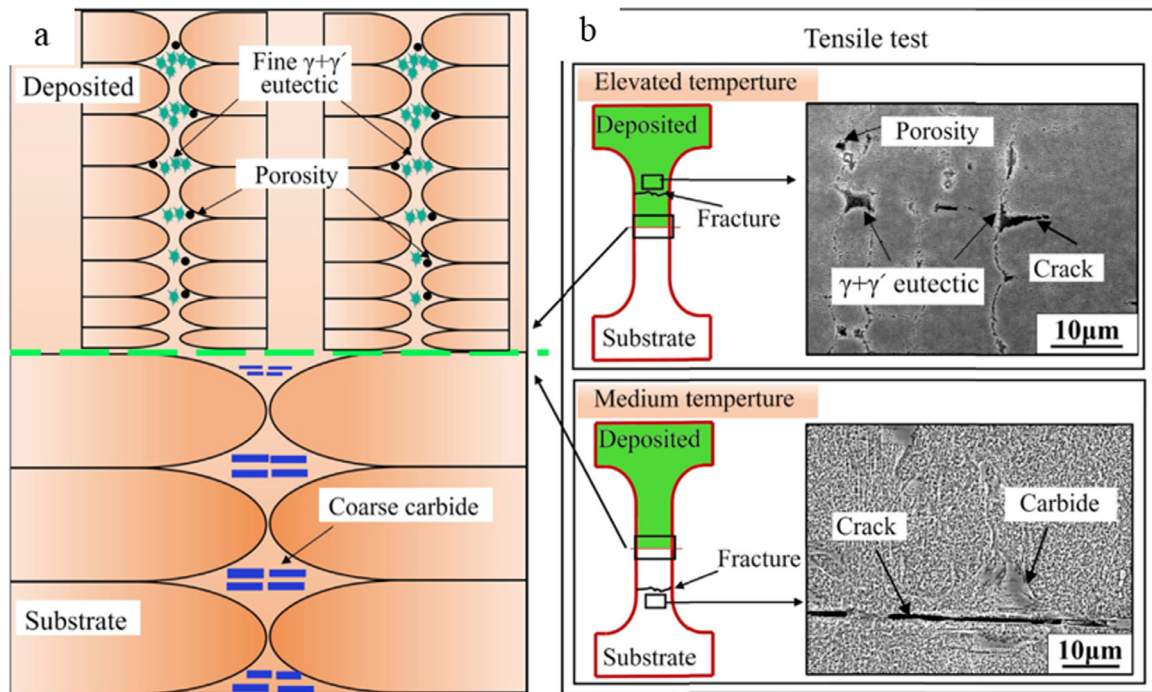


Fig. 11. (a) Schematic of the evolution of the cooling rate and volume fraction of γ - γ' eutectic and (b) illustration of the distribution of the γ - γ' eutectic with the increase of deposited height [27].

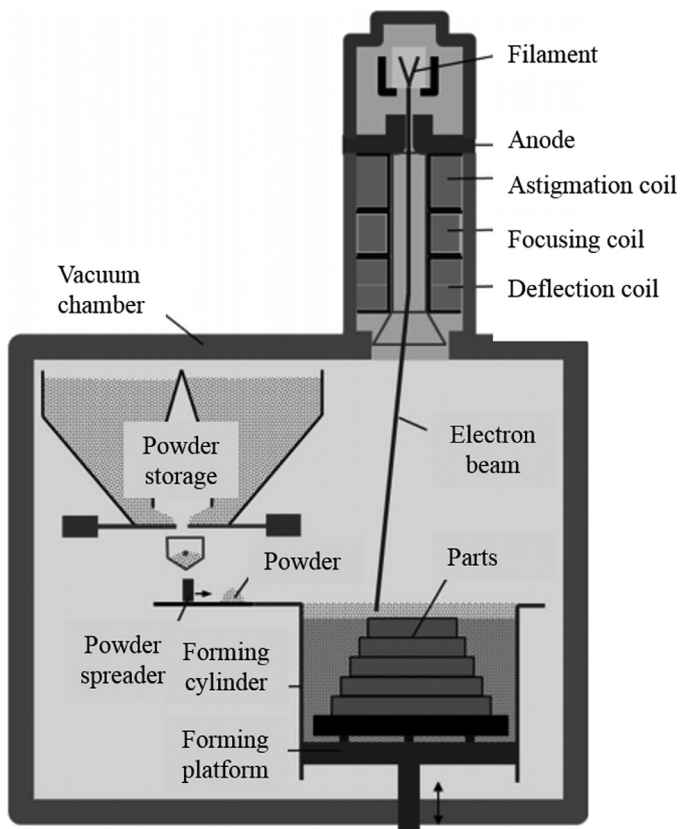


Fig. 12. Schematic illustration of electron beam powder bed fusion (EB-PBF) principle [37].

Patxi et al. [38] used an experimental method (MaxPro design) to study the SX formability of four superalloys: Nimonic 105, Haynes282, IN738, and MarM-247. Three experimental parameters were controlled to control the interval, line offset, speed function, and reference current. They found that higher energy configurations can promote the growth of SXs. Except for MarM-247, in which cracking was observed, all three other materials obtained a crack-free SX structure. Among them, Nimonic 105 showed the largest SX processing window, as shown in Fig. 14. In Fig. 14, the build direction is vertical in all images. The out-of-plane direction is parallel to the line scan direction and printed sample cube face. Images sorted with respect to the surface energy density. And each image is 2 mm $\times$ 2 mm.

Li et al. [13] fabricated SX blocks of two different sizes, as shown in Fig. 15; this is the necessary approach for investigating the adaptations for variant part sizes. As the growth of an SX is very sensitive to the solidification condition, changing the length of scan line would alter the temperature of the molten pool, thereby changing the solidification condition. The parameter used for the 15-mm-wide specimens was not suitable for the 20-mm-wide specimen when fabricating an SX. The processing window for fabricating an SX is narrow and scan length-dependent, by which SX bars are fabricated in the core of the specimens. In the study of Li et al., the width of the SX bar was 11 mm in the 15-mm-wide specimens and 14 mm in the 20-mm-wide specimens, with the 2–3 mm cortex regions being polycrystalline. Like Edouard et al. [12], Patxi et al. [38], and Li et al. [13] fabricated SX blocks without a triangular structure designed in the bottom of the specimens, proving that a triangular structure might not be necessary for fabricating SXs for superalloys.

3.2. Microstructure of superalloy SX built by electron beam powder bed fusion

The dendrite structures of the SX regions in the specimens of both widths are shown in Fig. 16. The primary dendrite arm spacing (PDAS) was experimentally measured as 15 μm by Li et al. [13], whereas the

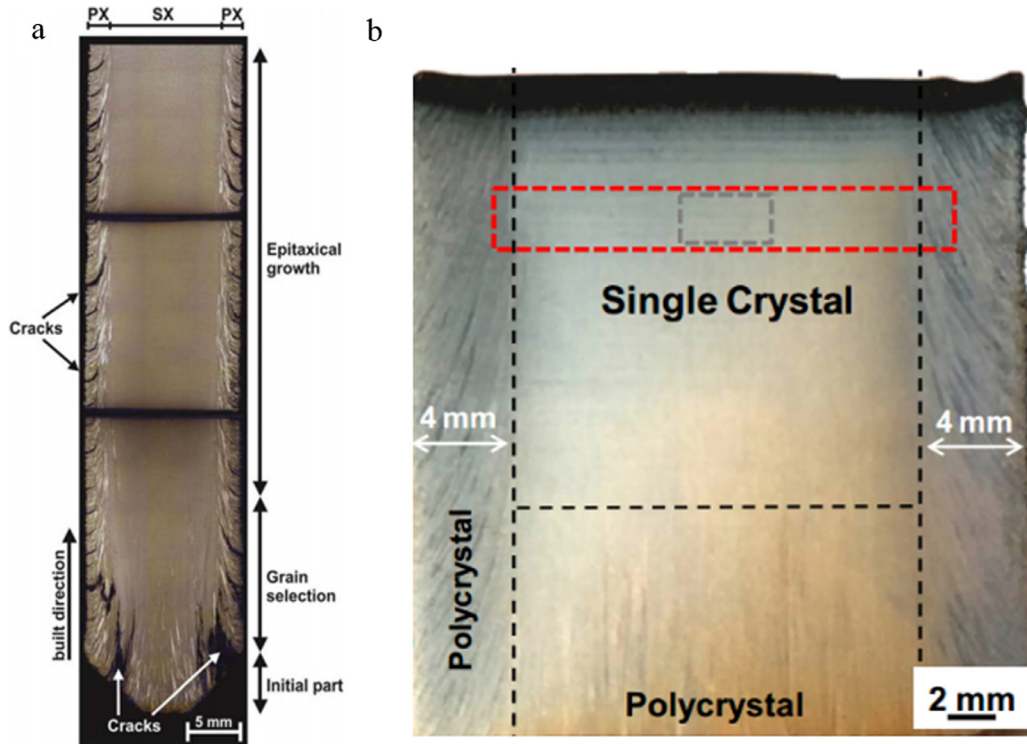


Fig. 13. Single crystal sample prepared by EB-PBF: (a) Ramsperger et al. [11]; (b) Edouard et al. [12].

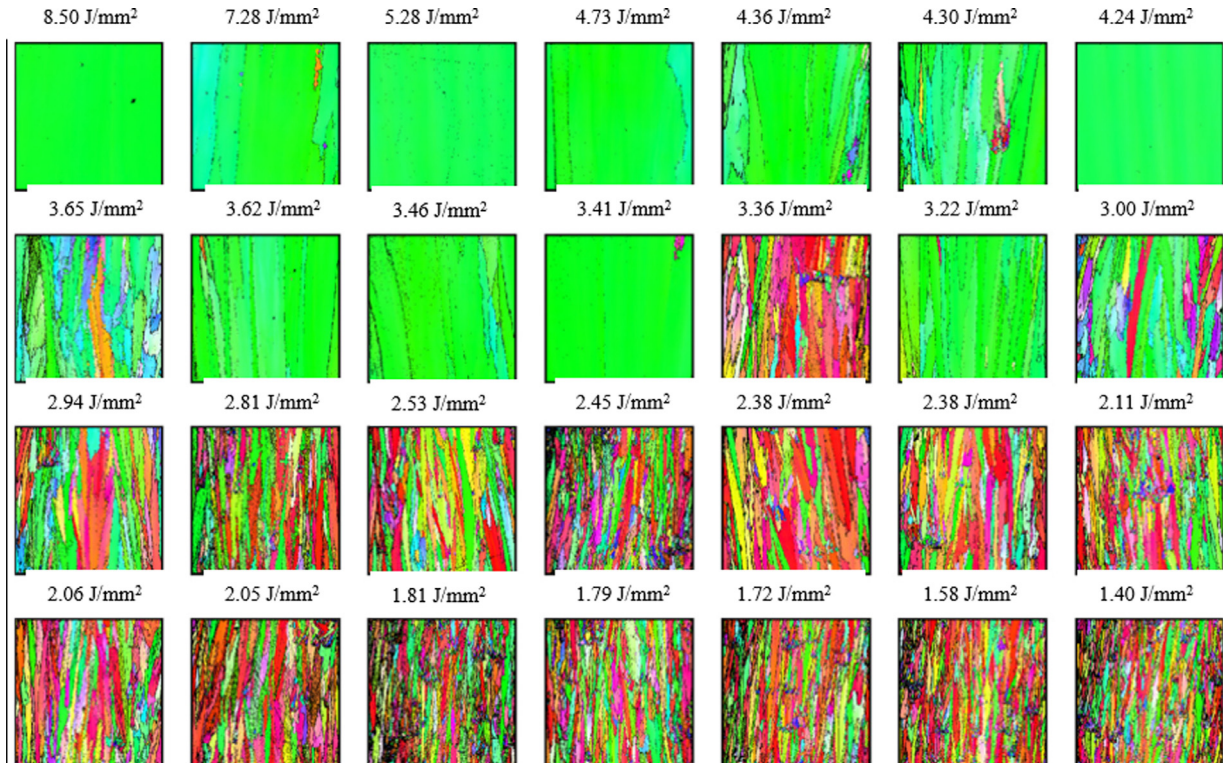


Fig. 14. Nimonic 105 inverse pole figure (IPF) maps visualizing the out-of-plane direction preference [38].

reported PDAS for the conventional high rate solidification and liquid metal cooling processes for an SM was approximately 200–400 μm [39, 40]. A PDAS of 200–400 μm indicates a cooling rate of ~ 0.1 – 1°C/s , whereas the PDAS of 15 μm indicates a cooling rate of approximately 200°C/s [39]. Thus, the cooling rate of the EB-PBF process was

higher than that of the casting process during the fabrication of an SX, as reported by Chauvet et al. [12]. Additionally, the carbide precipitates along the dendritic boundary and γ' phases could be found in as-built samples [13], and were much finer than those of a cast sample.

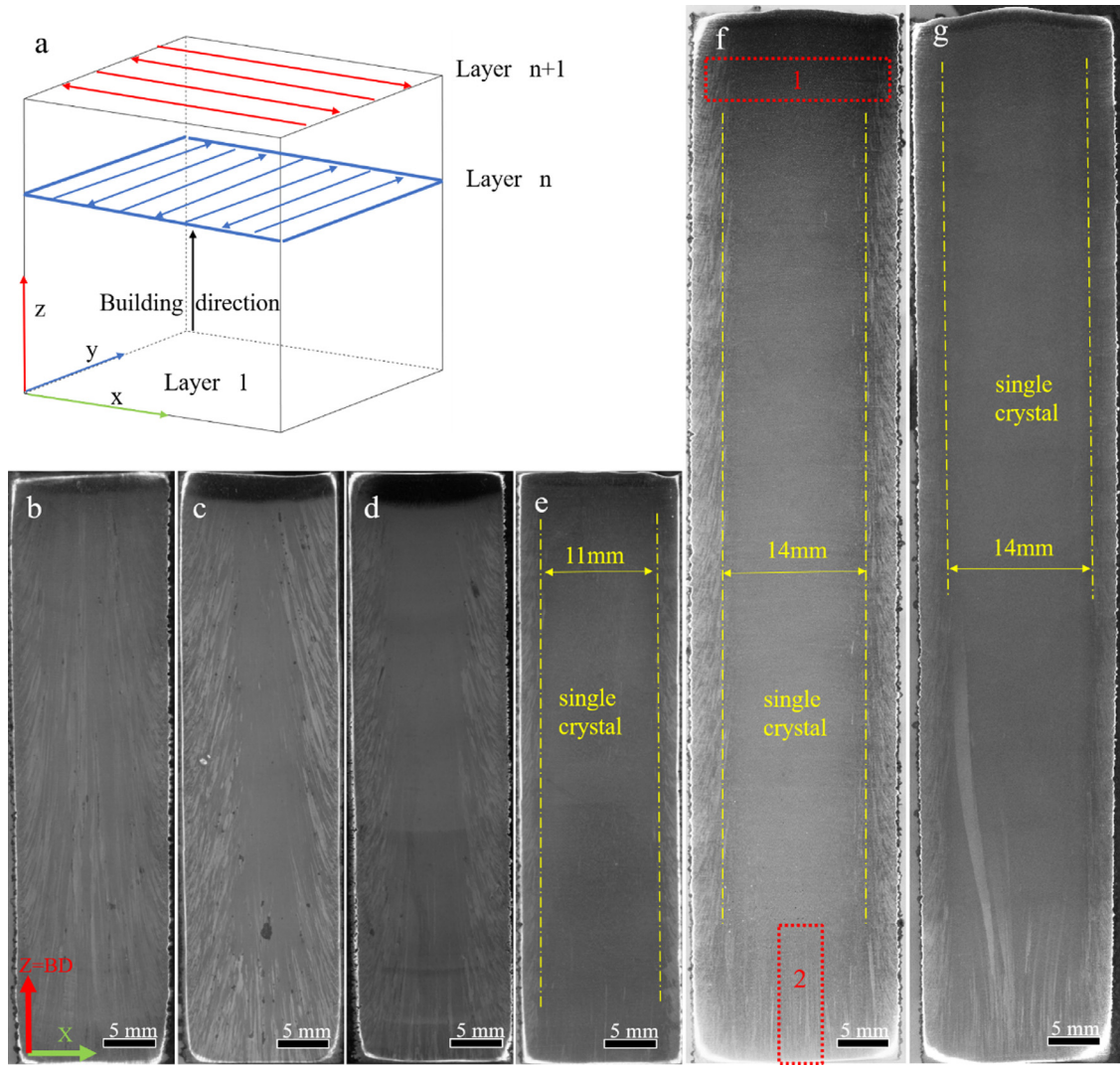


Fig. 15. Scanning strategy and stereoscopic photos of the defect-free specimens: (a) Scanning strategy; (b) 15 mm, $v = 3$ m/s; (c) 15 mm, $v = 2$ m/s; (d) 15 mm, $v = 1$ m/s; (e) 15 mm, $v = 0.6$ m/s; (f) 20 mm, $v = 0.8$ m/s; and (g) 20 mm, $v = 0.7$ m/s [13].

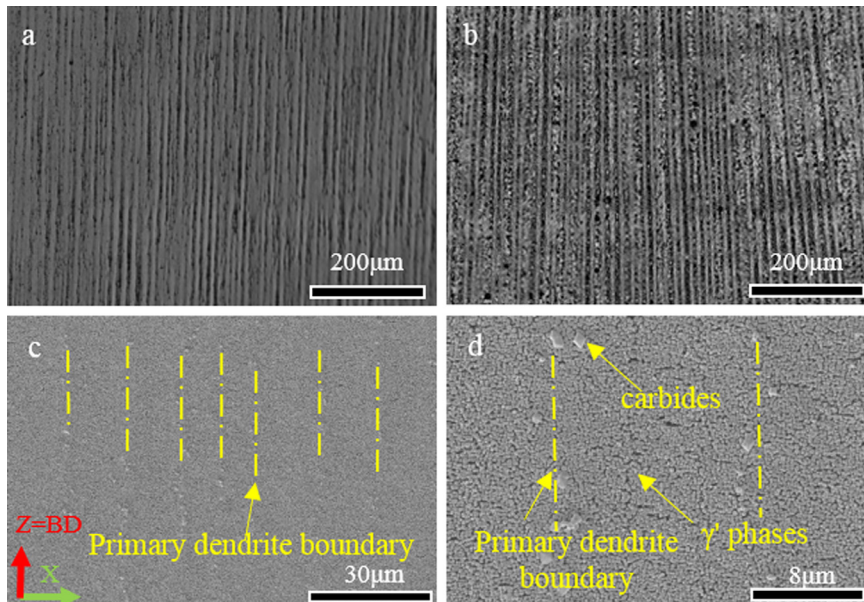


Fig. 16. Primary dendrite arm spacing (PDAS) of the samples: (a) OM of Fig. 15(e) approximately 15 μm ; (b) OM of Fig. 15(f) approximately 15 μm ; (c) SEM of Fig. 15(e); (d) high magnifications SEM of Fig. 15(e) [13].

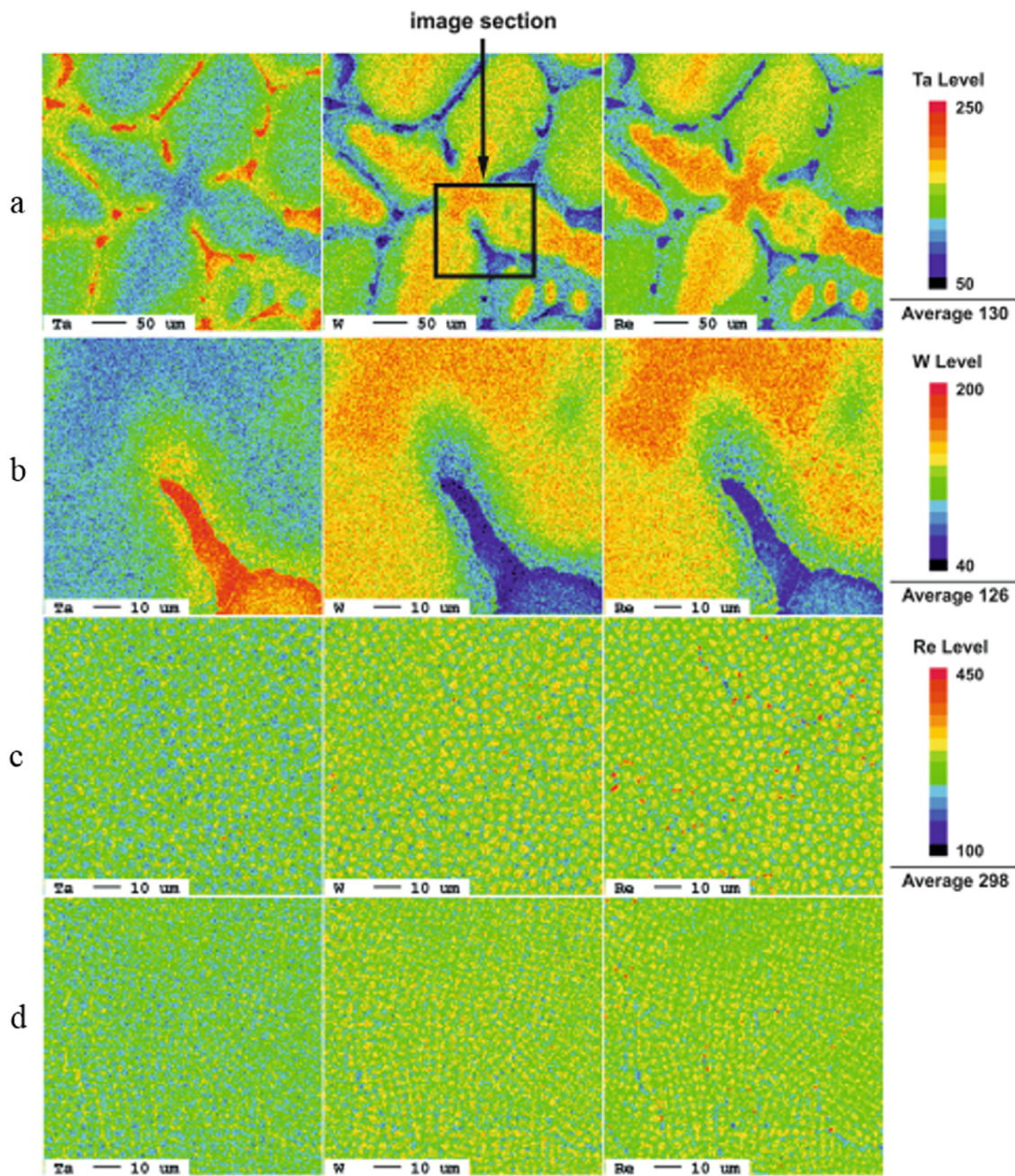


Fig. 17. Qualitative microprobe mappings (horizontal cross section) of Ta, W, and Re: (a) as-cast sample analyzed with a spot size of $2\ \mu\text{m}$; (b) Magnification of (a) (spot size $0.5\ \mu\text{m}$); (c) EB-PBF sample ($P = 510\ \text{W}$, $v = 6000\ \text{mm/s}$, line offset = $0.03\ \text{mm}$) (spot size $0.5\ \mu\text{m}$); (d) EB-PBF sample ($P = 180\ \text{W}$, $v = 300\ \text{mm/s}$, line offset = $0.1\ \text{mm}$) (spot size $0.5\ \mu\text{m}$) [40].

Ramsperger et al. [40] compared the dendrites, strengthening phase size, and chemical composition (Ta, W, and Re content) of casting and electron beam AM (shown in Fig. 17). The chemical compositions of the crack-free CMSX-4 SX samples built by EB-PBF were highly homogeneous, and two orders of magnitude smaller than that of traditional castings. This had a great influence on the size distribution of the precipitated phase, and increased the solidus temperature by 16 K. Therefore, they believed that EB-PBF was very promising for producing defect-free and complex-shaped parts for applications demanding such remarkable microstructures and properties.

A hot isostatic press (HIP), a common post-treatment for high-performance AM parts, was used in the work of Rutttert et al. [41]. The effects of the constant pressure of the HIP with various HT temperatures and times on the microstructure were studied, and are shown in Fig. 18.

The results show that only a few minutes of dissolution are needed to dissolve the segregation and close the pores sufficiently; this is regarded as an advantage of the initial uniformity of the chemical composition in the parts built by EB-PBF.

To understand the evolution and dynamic behaviors of the γ' phase during the EB-PBF AM process, Wahlmann et al. introduced the synchrotron-radiation-based small-angle X-ray scattering (SAXS) method, aiming to investigate the growth and coarsening kinetics of γ' precipitates in CMSX-4, with in-situ heating and quenching to simulate the thermal cycles in the EB-PBF process [42]. It was concluded that the γ' phase precipitates extremely fast from the homogeneous nucleation, and its fraction remains below the equilibrium fraction throughout the process. The γ' phase coarsens over a timescale of several seconds, and a fraction of the γ' phase that was dissolved during heating re-precipitates

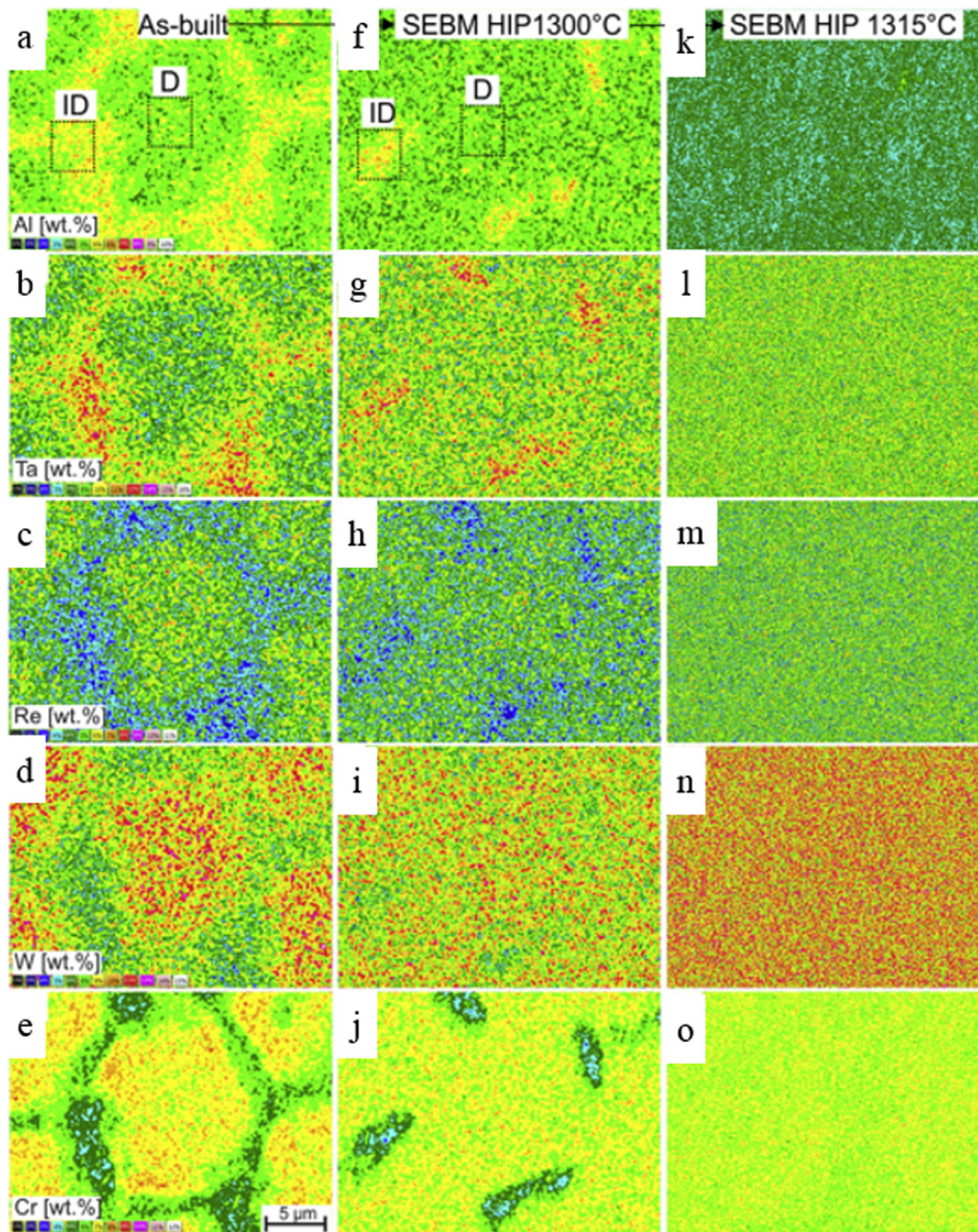


Fig. 18. Energy dispersive X-ray spectroscopy (EDS) element mappings of the EB-PBF material built with a line energy of 0.72 J/mm in the as-built state (a–e) and after two (f–j and k–o) different subsequent hot isostatic press (HIP) treatments each for 4 min holding time at the target temperature (the different elemental partitioning in the dendritic (D) and inter-dendritic (ID) regions is clearly discernible) [41].

by forming new particles, instead of growing on already existing precipitates. The morphology of the γ' phase changes during the thermal cycles (shown in Fig. 19); it is likely shaped as rounded cubes when precipitating from 1300°C, and as shape-edged cubes when re-precipitating from relatively low temperatures.

The TCP phase formation in the cast material is inhomogeneous, owing to the strong Re segregations in the homogenized state. Pistor et al. [43] investigated the homogeneity formation kinetics of TCP phases in a CMSX-4 SX built by the EB-PBF process, as shown in Fig. 20. A fast formation of TCPs is observed within the dendrite cores, whereas no TCPs form within the inter-dendritic areas of the AM superalloys. The maximum local TCP area fraction is strongly reduced compared to the cast state. The homogeneity of the AM SX is much higher than that of SXs produced by an investment casting process.

3.3. Mechanical properties for superalloy SX built by electron beam powder bed fusion

From the description in Section 3.2, it can be concluded that the SX microstructures of the EB-PBF process are different from those resulting from the traditional process. The mechanical properties of SX EB-PBF parts are sensitively affected by the microstructures, which may lead to particular characteristics. The tensile properties [44], creep properties [45], fatigue lifetime [46], and oxidation resistance [47] of EB-PBF CMSX-4 alloy samples were studied by Prof. Körner's group.

The tensile properties of these SX EB-PBF CMSX-4 alloy samples were compared to a cast sample by Körner et al. [44]. The yield strengths and ultimate tensile strengths of the SX EB-PBF samples in the as-built and heat-treated states at variant temperatures (from room temperature to

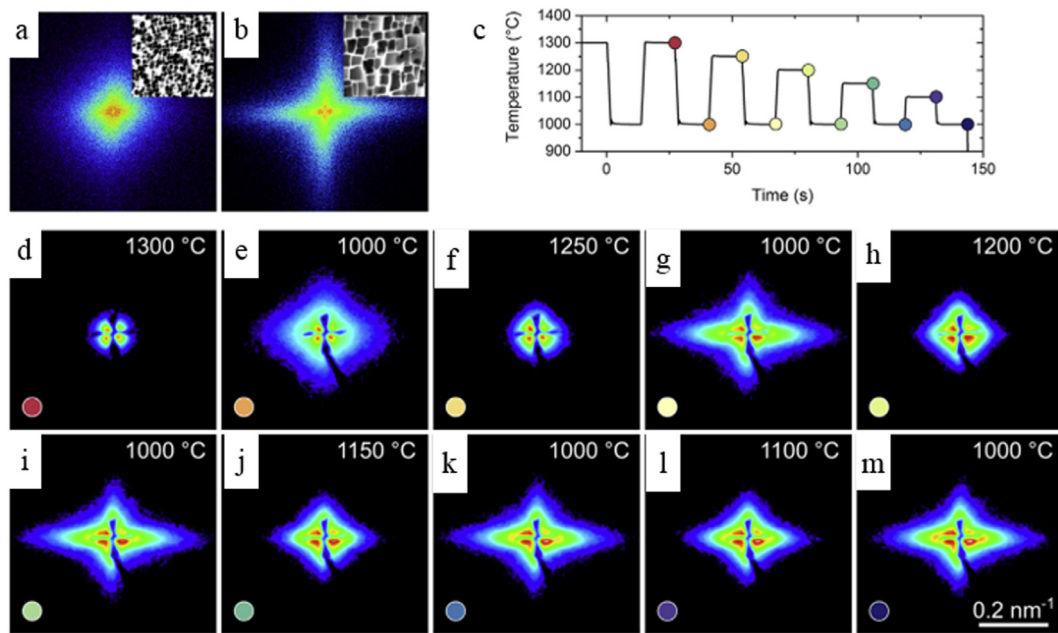


Fig. 19. Evolution of the scattering pattern during thermal cycling with 1000°C base temperature: Panels (a) and (b) show Fourier transforms of the microstructures. The round-edged cuboids create a diamond-shaped scattering pattern (a), whereas the sharp-edged precipitates yield a cross-shaped one (b). Panel (c) indicates the position of the patterns in the temperature diagram with colored dots. The individual patterns (d)–(m) evolve from diamond-shaped to cross-shaped at the lower temperature stages, illustrating the changes in the precipitate morphology from rounded cubes to sharp-edged cubes. The patterns are anisotropic, owing to the regular arrangement of the precipitates. The cross-shaped intensity dip in the middle of each pattern is the shadow of the beamstop [42].

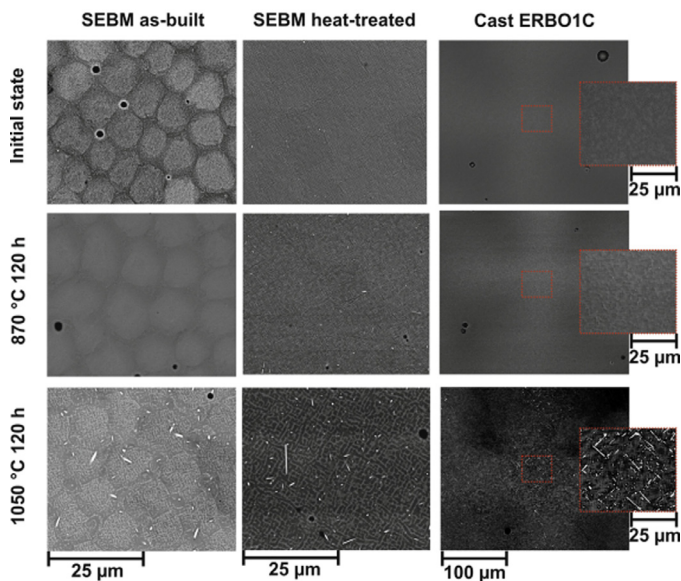


Fig. 20. Microstructure of the investigated material states EB-PBF as-built (left column), EB-PBF heat-treated (middle column) and cast ERBO1C (right column) in the initial state (top row) and after 120 h annealing at 870°C (middle row) and 1050°C (bottom row) [43].

950°C) had similar values to the cast sample (see Fig. 21). There was no significant difference between the EB-PBF as-built and HT samples. This behavior results from the pronounced in situ HT characteristic for the EB-PBF, as supported by the fine microstructure owing to the rapid solidification.

The [001] tensile creep properties, an important performance metric for an SX superalloy tested in the high-temperature/low-stress and low-temperature/high-stress creep regimes for three material states (EB-PBF, EB-PBF + HT, and Bridgman processed + HT), were studied by

Bürger et al. [45]. The results showed that after an appropriate precipitation HT, the EB-PBF processed SX Ni-based superalloys show creep properties matching (in the high-temperature/low-stress creep regime (1050°C, 160 MPa)) or even outperforming (in the low-temperature and high-stress creep regime (800°C, 600 MPa)) those of conventionally cast SX materials (see Fig. 22). The larger rupture strains of the heat-treated selective electron beam melting material result in slightly longer creep lives in the low-temperature and high-stress creep regime.

The fatigue lifetime is another important metric for an SX superalloy. Meid et al. [46] found the fatigue lifetime of a heat-treated material was significantly higher than that of an EB-PBF material, as shown in Fig. 23. Applying HIP with an integrated HT (IHT) resulted in an even longer fatigue life. Lifetime-limiting crack initiations occurred at the interfaces of the melting layers, at the micro-porosity areas generated during solidification, or, in the HIP-treated samples, at the precipitates which formed at the location of collapsed pores. If the pores are closed in the HIP-IHT, the crack initiation sites are removed or replaced by intermetallic inclusions which frequently precipitate at the former pore locations. With the crack initiation sites becoming fewer and smaller, the distances between the defects increase as well; together, these explain the improvement of the fatigue lifetime owing to the HIP-IHT for both cast and EB-PBF materials. HIP-IHT is necessary for an EB-PBF CMSX-4 alloy to obtain good fatigue properties.

In addition, the good fatigue lifetime and creep properties at high temperatures, the oxidation resistance of EB-PBF SX superalloys is better than that of investment casting alloys. Pistor et al. [47] noticed that the element segregation is strongly inhibited during the fast-cooling in EB-PBF. A time-resolved isothermal gravimetric analysis was thus conducted at 850°C and 1050°C in synthetic air for 100 h for AM and cast samples. Because persistent solidification-induced segregations (dendrites) accelerated the oxide growth, the homogeneity of the element distribution in the EB-PBF sample reduced this negative impact on the oxidation resistance, whereas the cast samples suffered (see Fig. 24).

In summary, the EB-PBF technology breaks the convention that SX seeds or grain selectors are required for the fabrication of SXs sam-

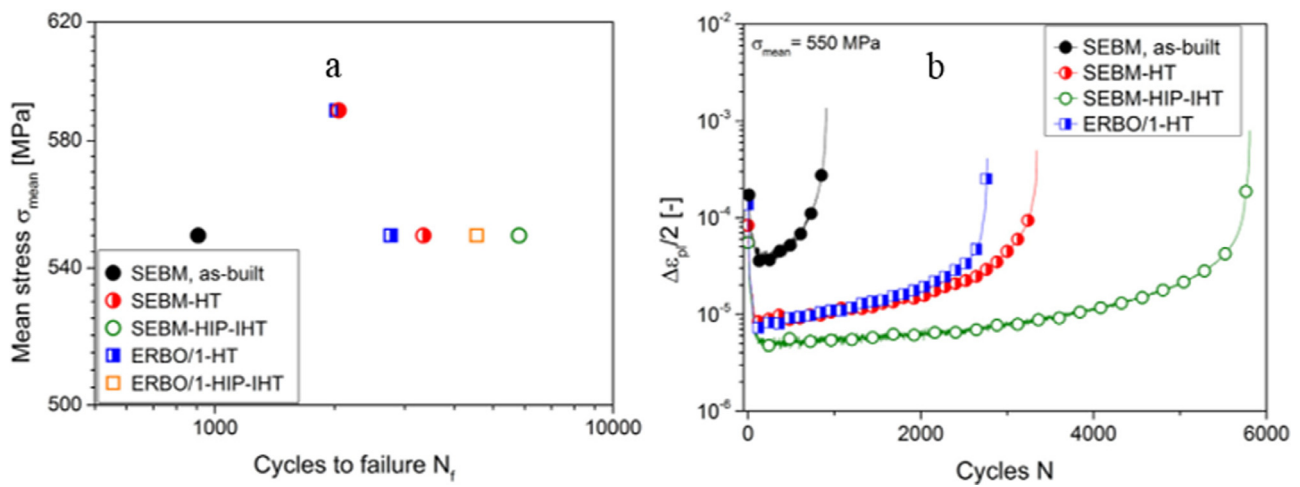


Fig. 23. Results of low cycle fatigue (LCF) experiments on differently heat treated specimens: (a) lifetime data summarized in a Wöhler-type diagram and (b) evolution of the plastic strain amplitude as function of cycle number [46].

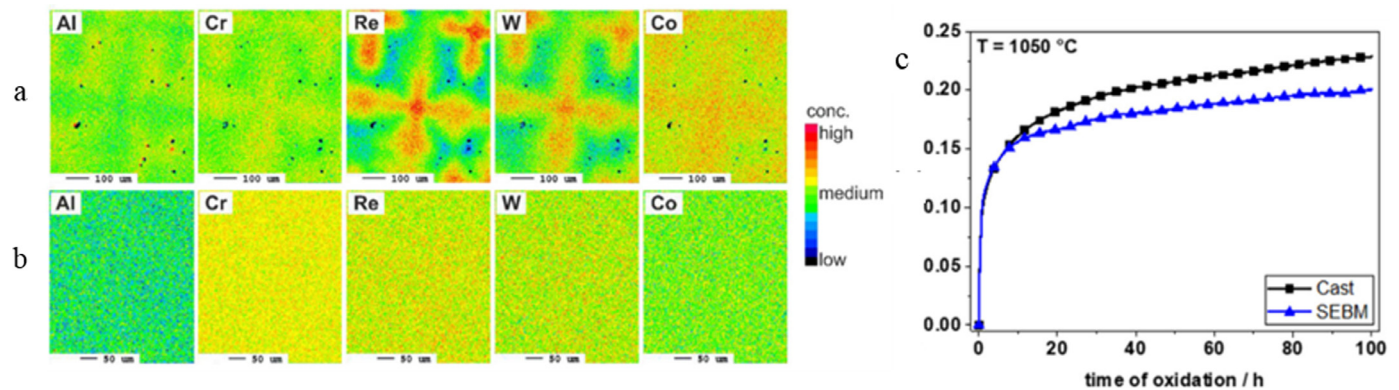


Fig. 24. Elemental distribution maps determined by EPMA with the respective elements denoted within the micrographs: (a) cast-CMSX-4 after heat-treatment; (b) EB-PBF CMSX-4 after heat-treatment; (c) continuous specific mass-gain for cast-material and EB-PBF-material oxidized for 100 h at 1050°C [47].

et al. [50] proposed an interdependence theory to discuss the relationship between the grain growth and nucleation in front of the dendrites, i.e., the CET. In this theory, when moving away from a growing grain, the undercooling in the liquid increases as the solute concentration gradually changes towards the nominal solute concentration. When the undercooling is higher than the minimum undercooling necessary for the nucleation, nucleation occurs, thereby hindering the growth of the existing grain. This theory was adopted in AM to investigate the impacts of the temperature gradient and solidification velocity on the nucleation in front of epitaxially growing grains [51–53]. Plots regarding the relationships of solidification condition and grain structure are shown in Fig. 25 [51].

As shown in Fig. 25, a higher temperature gradient and lower solidification velocity are preferred for inhibiting nucleation and thus promoting a grain's epitaxial growth. This phenomenon was also revealed by Ref. [53] with numerical modeling. A single track is the fundamental unit of AM. In single-track experiments and simulations, researchers also found more nucleation at the top region of the molten pool [51]; this was attributed to the lower temperature gradient and higher solidification velocity in the top region. However, the top regions are usually removed when depositing the next layer, and thus would not affect the epitaxial growth of the existing grains. In contrast, nucleation at the molten pool bottom [54] (see Fig. 26) has a greater potential to become valid new grains, and thus results in an equiaxed grain structure in the as-built components. Ding et al. [54] believed that the nucleation at the molten pool bottom is caused by the moving of the combination of the

temperature gradient and solidification velocity towards the equiaxed zone (see Fig. 25) owing to the elevated preheating temperature (1298 K). In another experiment by Ding et al. where no preheating was exerted, no nucleation was found in the molten pool bottom, regardless of scanning speed.

According to the temperature gradient and solidification velocity, Birmingham et al. [52] pointed out the positions of different AM technologies, as shown in Fig. 25. SLM usually has the highest temperature gradient and highest solidification velocity, owing to the low powder bed temperature and relatively small beam diameter. As shown in Fig. 25, it is located at the right-top corner, indicating a great potential for the fabrication of SX parts. However, there are almost no reports regarding the successful fabrication of SX bars using SLM. In contrast, LMD and wire arc AM (WAAM) usually have a small scanning velocity and large beam size, leading to a significantly reduced temperature gradient and solidification velocity (see Fig. 25). With these two AM technologies, growths of relatively large SXs are successfully fabricated [20, 25]. Furthermore, in EB-PBF, the powder bed temperature is usually very high, and the scanning speed is of the same magnitude as SLM. The temperature gradient would be much lower than that in SLM. According to the numerical modeling works [34, 55], the temperature gradient for a nickel-based superalloy in EB-PBF is approximately 10^4 – 10^6 K/m, and the solidification velocity is approximately 10^{-3} – 10^0 m/s. The actual range for each experiment is definitely processing parameter-dependent. For instance, in Ref. [12], where a centimeter-scale SX bar is produced, the low solidification velocity is deduced based on the measured PDAS

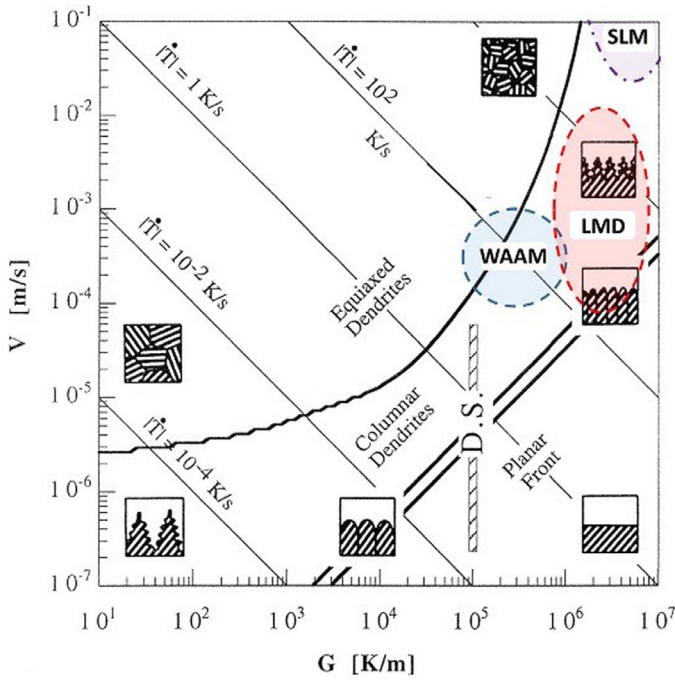


Fig. 25. Relationship between temperature gradient (G), solidification velocity (V), and obtained grain structure [51].

(approximately 10^{-2} m/s). These values mean that the EB-PBF should be located at the region close to the WAAM zone, and might be at the upper region of the WAAM. As discussed in Section 3, several research groups have succeeded in fabricating centimeter-scale SX bars using EB-PBF.

That is, although all AM technologies show the possibility of suppressing the nucleation of grains, fabricating large-scale SX bars from a non-SX substrate remains challenging, and only a few of the AM technologies (such as EB-PBF) have succeeded in the fabrication of SX bars. Therefore, there are additional mechanisms beneath the AM SX-fabrication besides maintaining the epitaxial growth of the grains.

4.2. Grain growth process

In directional solidification casting, which is usually used for manufacturing SX parts, a grain selector (see Fig. 1(a)) is adopted to choose a grain with an orientation parallel to the withdrawing direction, by geometrically blocking the growth of grains with un-preferred orientations. Gotterbarm et al. (see Fig. 27(b)) believed that the grain selection in an EB-PBF process where the SX bars are successfully fabricated should be similar to the process in directional casting. The grain selector is achieved by tailoring the preferred solidification direction; this is obtained by changing the scanning direction in the manner shown in Fig. 27(a).

The solidification direction in each layer is believed determined by the shape of the molten pool, especially the region at the bottom of the molten pool (see Fig. 28), i.e., the normal direction of the molten pool boundary at the bottom region. As discussed in Refs. [49, 56], the solidification direction varies with the processing parameter and scanning direction. With a certain solidification direction, the growth of grains with orientations close to this direction would be promoted owing to their superiority in grain growth competition, as given by the nature of the crystal. At that time, grains with orientations deviating from this direction would be suppressed.

Gotterbarm et al. [56] indicated that the grains deviating from the building direction can be gradually removed by changing the solidification direction in a certain order; this can be achieved by rotating the scanning direction by a degree after completing each layer. Successful practices in Refs. [12, 38, 56, 57] all demonstrated that rotating the scanning direction by 90° is promising for the fabrication of SX parts (see Fig. 27(a)). Gotterbarm et al. [56] believed that the rotation of the scanning direction actually causes a change in the solidification direction as shown in Fig. 27(b), which mimics the grain selector in directional casting. In contrast, in AM, the grains with unpreferred orientations are removed, owing to the loss of grain growth competition.

Interestingly, Fernandez et al. [38] also fabricated SX bars by rotating by 180° . They also found that growth of the SX was disrupted when the scanning direction suddenly changed by 45° .

Fernandez explained the special shape of the molten pool. In his hypothesis, if the molten pool is sufficiently short and wide, a trailing tear-drop will form an angle of approximately 45° , thereby yielding co-incident crystal lattices consistent with the preferred orientation [001] along the building direction over many layers (see Fig. 29). As a result,

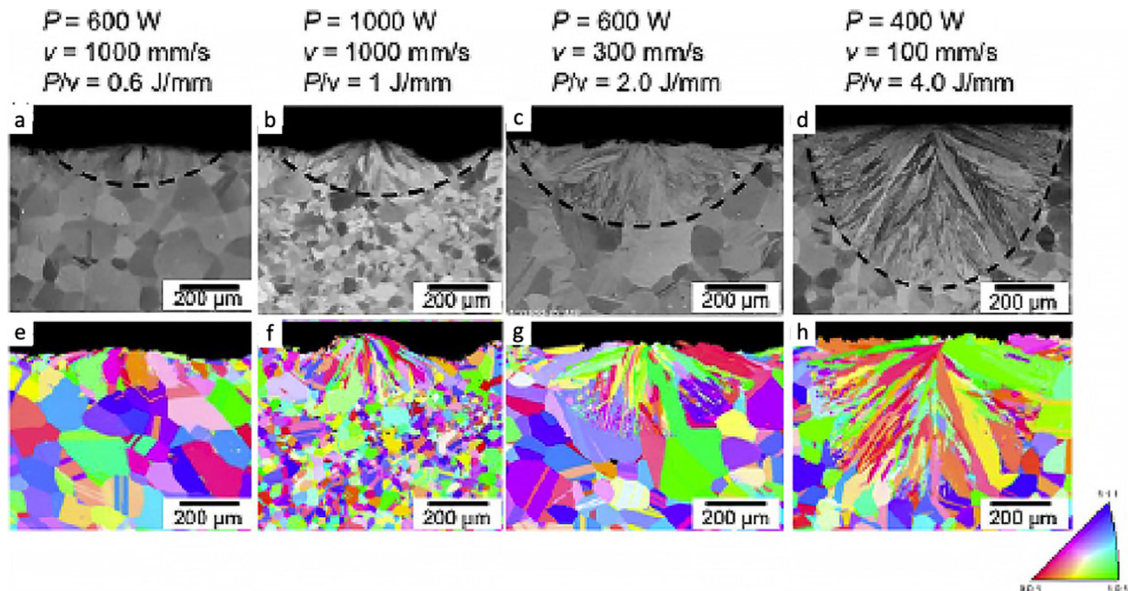


Fig. 26. Single-track experiments using electron beam selective melting with preheating temperature of 1298 K [54].

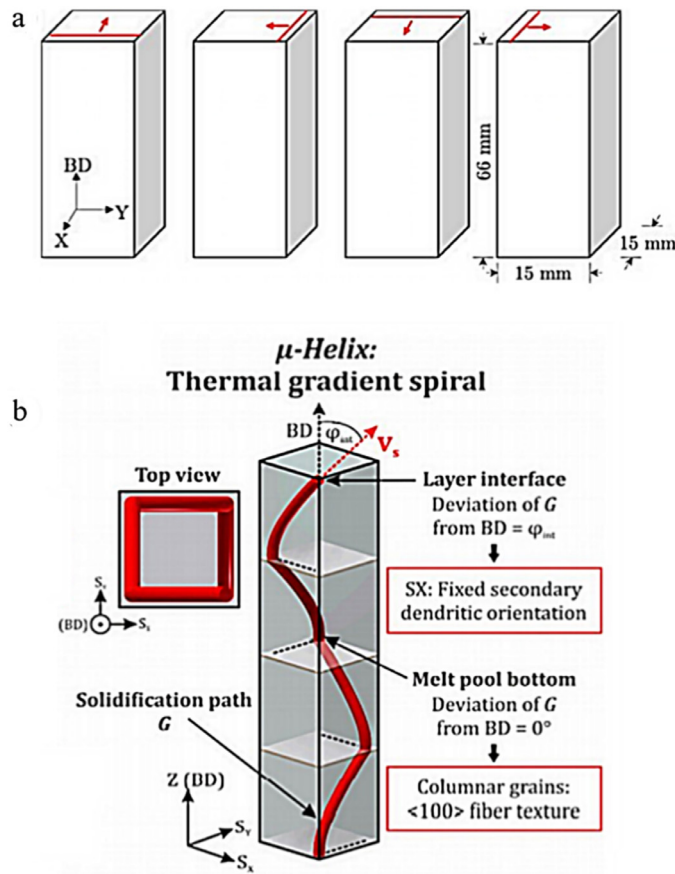


Fig. 27. (a) Schematic diagram of the scanning strategy, where the scanning direction rotates by 90° after finishing the previous layer [54]; (b) grain selection route proposed by Ref. [56].

the grains with preferred orientation could be selected only by rotation of scanning direction by $180/90^\circ$. However, his simulation results do not support this theory.

In contrast to the theory proposed in Refs. [38, 56], where only the solidification direction in an independent melting track is accounted for, Pistor et al. [57] emphasized the importance of the interactions of adjacent tracks. In the cases where the SX was successfully fabricated, a persistent molten pool is observed, indicating that the molten pool remains unconsolidated when starting scanning the second, third, and even later tracks. These tracks constitute a much larger molten pool whose direction is perpendicular to the scanning direction, and whose speed is named the lateral velocity ($v_{lateral}$). Fig. 30 shows a processing map related to the lateral velocity. It shows that only a narrow range of $v_{lateral}$ is suitable for SX fabrication.

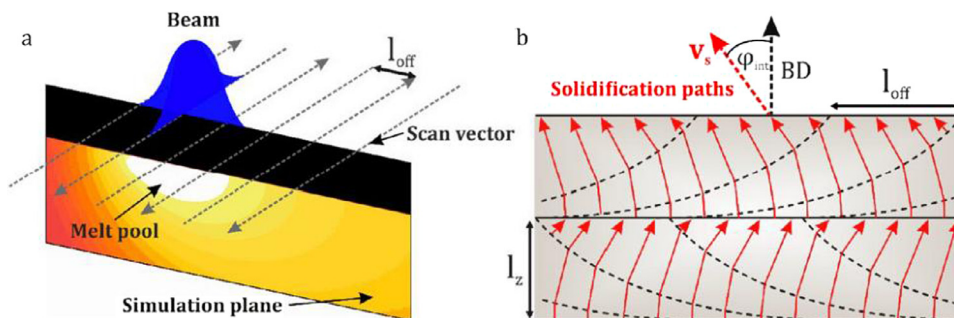


Fig. 28. (a) 2D simulation result of the molten pool; (b) Solidification direction of the adjacent two layers extracted from the 2D simulation [56].

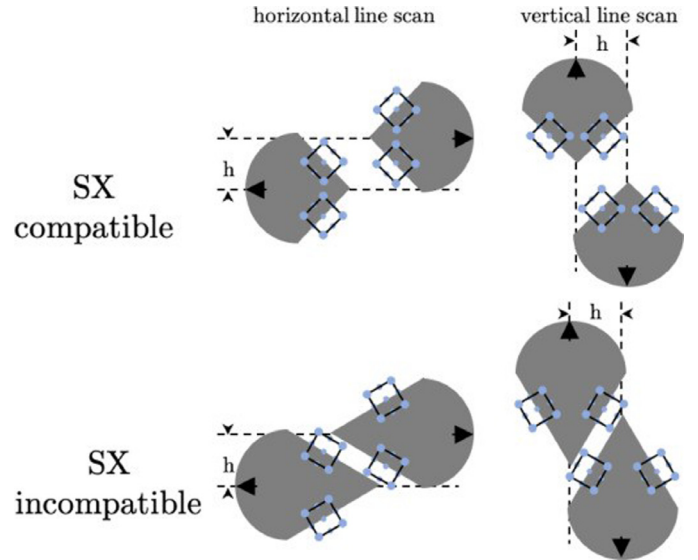


Fig. 29. Grain selection mechanism proposed by Fernandez-Zelaia et al. [38].

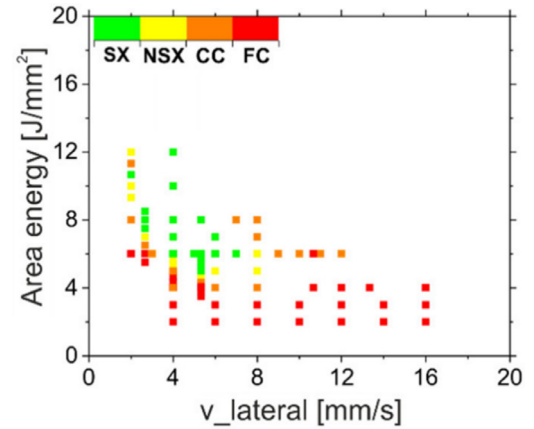


Fig. 30. Processing map of SX fabrication via EB-PBF [57].

Both Gotterbarm et al. [56] and Pistor et al. [57] believe that controlling the angle between the solidification orientation and building direction is critical. In Ref. [57], the suitable angle between the solidification orientation and building direction ranged from 12° – 17° . In Ref. [56], the measured maximum angle was 26° for the case with the emergence of the SX. This angle is out of the range in Ref. [57], and could be owing to the varied evaluation method. The angle was measured considering the multi-track interaction by Pistor et al. [57] but only for in a single track by Gotterbarm et al. [56]. Furthermore, Pistor et al. [58] found that the orientations of the secondary dendrite arms in the as-built SX bar

Table 1
The strategies in the trial of SX fabrication and repair using AM.

Process	Materials	Dendrite width	SX mechanism	SX size (approx.)	Ref. No.
L-PBF	SRR99	1–2 μm	Induced grain growth from SX substrate	$6 \times 4 \times 6 \text{ mm}^3$	[28]
DED	SRR99 and René N5	Several microns	Induced grain growth from SX substrate	2 mm width	[22]
DED	CMSX-4	—	Induced grain growth from SX substrate	$10 \times 5 \times 2.5 \text{ mm}^3$	[23]
DED	CMSX-4	11 μm	Induced grain growth from SX substrate	—	[59]
DED	CSU-B1	Several microns	Induced grain growth from SX substrate	$10 \times 10 \times 1.5 \text{ mm}^3$	[60]
DED	CSU-B1	$\sim 5 \mu\text{m}$	Induced grain growth from SX substrate	$10 \times 10 \times 5 \text{ mm}^3$	[25]
DED	CMSX-10	—	Induced grain growth from SX substrate	$20 \times 1.2 \times 2 \text{ mm}^3$	[26]
DED	DD32	17.2 μm	Induced grain growth from SX substrate	$44 \times 18 \times 3 \text{ mm}^3$	[27]
DED	DD32	—	Induced grain growth from SX substrate	$28 \times 17 \times 2 \text{ mm}^3$	[61]
DED	DD98M	Several microns	Induced grain growth from SX substrate	$50 \times 50 \times 80 \text{ mm}^3$ (Tubular with thickness 10 mm)	[62]
DED	DD6	Several microns	Induced grain growth from SX substrate	$30 \times 5 \times 2 \text{ mm}^3$	[63]
DED (quasi-continuous-wave laser)	IN718	Several microns	(From SS substrate) Induced grain growth from SX substrate	Only SX-like	[24]
EB-PBF	IN718	—	Grain selection	$5 \times 5 \times 10 \text{ mm}^3$	[64]
EB-PBF	IN738	15 μm	Grain selection	$14 \times 14 \times 95 \text{ mm}^3$	[13]
EB-PBF	IN738	—	Grain selection	$2 \times 2 \times 2 \text{ mm}^3$	[38]
EB-PBF	Haynes 282	—	Grain selection	$1 \times 2 \times 2 \text{ mm}^3$	[38]
EB-PBF	MarM-247	—	Grain selection	$0.5 \times 2 \times 2 \text{ mm}^3$	[38]
EB-PBF	Nimonic 105	—	Grain selection	$2 \times 2 \times 2 \text{ mm}^3$	[38]
EB-PBF	Ni-Co-Cr-Mo-Al-Ti-B	30 μm	Grain selection	$14 \times 14 \times 15 \text{ mm}^3$	[12]
EB-PBF	CMSX-4	10 μm	Grain selection	$8 \times 8 \times 70 \text{ mm}^3$	[64]
EB-PBF	CMSX-4	10 μm	Grain selection	$10 \times 10 \times 35 \text{ mm}^3$	[11]
EB-PBF	CMSX-4	10 μm	Grain selection	$12 \times 12 \times 10 \text{ mm}^3$	[57]
EB-PBF	CMSX-4	10 μm	Grain selection	$10 \times 10 \times 40 \text{ mm}^3$	[58]

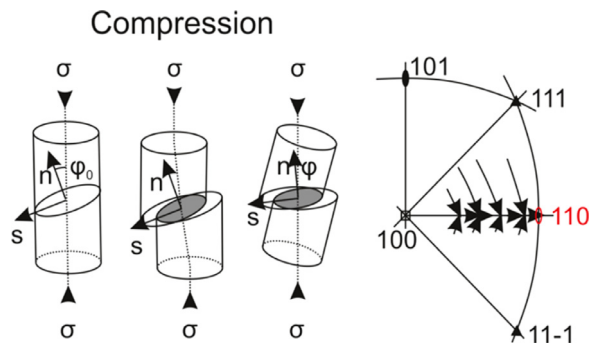


Fig. 31. Schematic drawing of the texture formation in face-centered cubic (FCC) crystals under compression that leads to distinct rotation towards [110] under compression [58].

are [110]. In their experiment, an SX substrate with secondary dendrite arms with the orientation [100] gradually changed to [110], indicating the existence of the orientation selection for the second dendrite arms. Pistor attributed this phenomenon to a grain twist caused by thermal compression stress (see Fig. 31). As the scanning strategy affects the compression stress, the orientation selection process can be tailored by changing the scanning direction.

From the literature, the temperature gradient, solidification velocity, and solidification orientation (or the evolution of the molten pool) have proven critical for SX fabrication, and the evident relationships between these factors and the possibility of SX fabrication has been revealed by

many researchers. However, when it comes to the actual AM process, there remains a need for a theoretical or empirical guideline to help identify the most potential processing window for fabricating SX components. Moreover, many questions need to be answered. For instance, there is a need to determine whether SX bar fabrication is feasible in ordinary SLM and why; similarly, there is a need to determine how can we minimize the grain selector and polycrystalline cortex regions in the as-built components, and whether it is possible to fabricate a SX hollow turbine blade. To answer these, further investigations on the fundamental mechanisms of SX fabrication in AM are needed.

5. Conclusions and Perspectives

As an emerging technology, AM has promising application prospects for the fabrication of superalloy SXs. At present, the relevant research can generally be divided into two aspects: laser AM and electron beam powder bed fusion. Among them, for the LMD and SLM technologies usually used for SX repair, an SX seed material is required to grow an SX through epitaxial growth. EB-PBF technology breaks the convention that SX seeds or spiral grain selection structures are required for the fabrication of SXs. The SX samples are fabricated directly through the grain selection scanning strategy, providing new ideas for the use of AM technology for SX fabrication.

The strategies for SX fabrication and repair using AM, as well as the relevant information, are summarized in Table 1. L-PBF or DED with metallic powder can only fabricate SX nickel-based alloys via induced grain growth from an SX substrate. Currently, only the EB-PBF can eliminate the use of an SX substrate through the dynamic grain selection mechanism, under a beneficial thermal environment.

At present, the challenges faced by the laser AM technology used in the fabrication of SXs are mainly the process methods and post-treatment, and the relevant mechanisms have been explained according to the epitaxial growth and the inhibition of valid nucleation. However, the mechanisms of the single-grain selection in the EB-PBF technology have not been clearly identified. An explanation of the grain selection process for a SX is one of the challenges for research in this area. A post-treatment coupled with LAM or EB-PBF processes should be optimized to obtain a fine microstructure and excellent mechanical properties for SXs.

Looking forward to the future, the author believes that because the EB-PBF technology does not use grain selectors or SX seeds to fabricate superalloy SXs, it may provide a new options for promoting the AM of superalloy SXs, even though the manufacturing of complex parts (such as hollow turbine blades) remains a challenge.

To overcome this challenge, a hybrid AM with a combination of an electron beam, laser, machining, or other processes might be a promising method and important direction for superalloy SX AM in subsequent research.

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CRediT authorship contribution statement

Yang Li: Conceptualization, Writing–original draft preparation, Investigation. **Xiaoyu Liang:** Writing–original draft preparation. **Yefeng Yu:** Writing–original draft preparation. **Dongfang Wang:** Validation. **Feng Lin:** Writing–review & editing, Project administration and supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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